

Abstract

In this essay, I will investigate how three different types of household detergents/soaps affect the surface tension of water.

I investigated three brands of laundry detergents, hand dish washing detergents and showering soaps. I predicted that laundry detergents would lower the surface tension of water the most because of their cleaning ability and their composition.

In a bid to measure the effect the detergents had on the surface tension of water, I used a personal adaptation of the Du Nuoy ring method, which enabled me to collect data that would allow me to compare the relative effect on the surface tension of water of each detergent/soap. The results I obtained showed that laundry detergents lowered the surface tension of water the most as the value of the force required to pull the circular metal from the surface of the water was lower compared to the force required to pull the circular metal from the surface of water respectively for hand dish washing detergents and showering soaps.

In conclusion, I claimed that it was possible that the ability of a detergent to lower the surface tension the water could be related to its cleaning ability or the composition of the surfactants. However, it is difficult to be certain since the full composition of the detergents are not publicly released.

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Introduction

Water is a chemical that covers a large portion of earth and is vital to life on earth. We use either soap or detergent for washing and cleaning everything from our clothes to our bodies. Saponification, one of the major processes by which soaps are created, constitutes of heating triglycerides together with aqueous alkali solutions which are commonly NaOH or KOH together to produce soap, water and glycerine (American Cleaning Institute “Soaps & Detergents: Chemistry”, n.d). Soaps are made of naturally occurring fats and oils from animals or plants (American Cleaning Institute “Soaps & Detergents: Chemistry”, n.d). On the other hand, detergent surfactants are made of petrochemicals and/or oleochemicals (American Cleaning Institute, “Soaps & Detergents: Chemistry (Surfactants)”. n.d).



Both soaps and detergents consist of molecules known as surfactants (University of York, 2003). Surfactant molecules have two ends (head and tail) (University of York, 2003). The head of the molecule is hydrophilic and is attracted to water (University of York, 2003). The tail is hydrophobic and is attracted to grease or dirt (University of York, 2003). Generally, for molecules to have these capabilities, they require an alkyl chain of at least 9 carbon atoms (Reusch, 2016). Surfactants are usually grouped into four types: anionic, non-ionic, cationic and amphoteric surfactants (CIEC Promoting Science, 2013).

Anionic surfactants have a negatively charged ion which is the hydrophilic part (CIEC Promoting Science, 2013). Cationic surfactants have a positively charged hydrophilic group (CIEC Promoting Science, 2013). Amphoteric surfactants contain hydrophilic groups

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that are both positively and negatively charged while the non-ionic surfactants do not contain an ionic hydrophilic group but their hydrophilic properties arise from containing oxygen atoms which can hydrogen bonds with the water molecules (CIEC Promoting Science, 2013). Non-ionic surfactants are more surface active and emulsifying than anionic when used in similar concentrations (CIEC Promoting Science, 2013).

Surface tension is formed in liquids when molecules on the surface cannot form bonds with molecules on all sides which causes the molecules to form stronger bonds with the molecules surrounding it (Nave, n.d). The intermolecular forces between H₂O are due to hydrogen bonding, which is also responsible for the high boiling point of water (Brown and Ford, 2014). However, when soaps/detergents are added in water the surfactant molecules float on the surface of water with the hydrophilic ends in the water and the hydrophobic ends away from the water (Reusch, 2016). As these molecules adsorb on the surface of the water, the forces of attraction between the water molecules and surfactant molecules is weaker than the forces of attraction between the water molecules, hence reducing surface tension (Atwood and Florence, 2008). Diminishing the surface tension of water allows the cleaning agents to penetrate the objects being cleaned (Milady, 2013).

Micelles form as the concentration of surfactants molecules in the water increases (Reusch, 2016). The hydrophobic ends are able to group together while avoiding the polar hydrophilic ends, thereby creating structures known as micelles (University of York, 2002). Since the outer part of a micelle consists of hydrophilic substances, which are polar, it allows substances such as grease to be washed away as they would be interacting with the hydrophobic ends inside the micelle (See Figure 2) (University of York, 2002). From figure 2 it can be seen that in terms of appearance, micelles are circular in structure, with the hydrophobic ends of the molecules concentrated in the inside of the circular structure while the hydrophilic ends form the outside of the structure.

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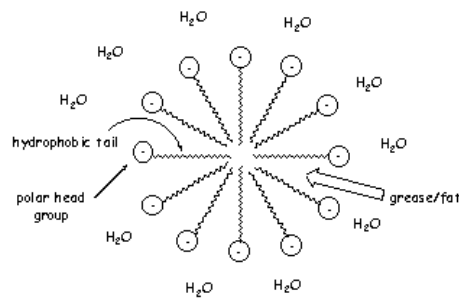


Figure 2: A Micelle -
<http://www.york.ac.uk/res/sots/activities/supermols.htm#>

Upon learning about surfactants, I was intrigued in determining whether the ability to lower surface tension varied between the types of soaps used. At times when washing clothes or dishes I noticed the water felt

more “slippery” after adding a detergent. For

this reason, I will investigate **how three types of**

household detergents affect the surface tension of water? I will test laundry

soaps/detergents, dishwashing soaps/detergents and showering soaps(gels) to determine

which ones lower the surface tension of water the most and to also see whether there is a link

between the cleaning ability of the soap/detergent and how much the surface tension is

lowered in the case of these three types of detergents. I believe doing this experiment would

provide information that can be useful to people that would need to know which detergent

would be the best for lowering the surface tension for the purpose of cleaning or just for

general knowledge.

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Exploration

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Hypothesis: Out of the three types of soap/detergents that I will be investigating, I predict that laundry soaps/detergents will lower the surface tension of water the most. Laundry soaps/detergents can utilise non-ionic surfactants and anionic surfactants (American Cleaning Institute “Soaps & Detergents: Surfactants & Builders”, n.d) (CIEC Promoting Science, 2013), out of these two groups of surfactants, non-ionic surfactants are known to be more surface active and to more emulsifying than anionic surfactants (CIEC Promoting Science, 2013). One advantage of having non-ionic surfactants is that they don’t interact with ions in hard water(CIEC Promoting Science, 2013). Laundry detergents also deal with the strong stains from clothes, if their cleaning ability is related to the surface tension; I believe they would require a low surface tension to function effectively. For these reasons, I believe that they will lower the surface tension of water the most.

Hand dishwashing soaps/detergents usually use anionic surfactants (American Cleaning Institute “Soaps & Detergents: Surfactants & Builders”, n.d). However, I believe that the combination of non-ionic and anionic surfactants that could be present in laundry detergents may be stronger so I believe that dishwashing detergents will lower the surface tension the second most. I believe showering soaps would have the lowest effect on surface tension as they use amphoteric surfactants, which are considered to be mild (CIEC Promoting Science, 2013). I believe that detergents/soaps that deal with tough stains would require a lower surface tension to have a strong cleaning action because, diminishing surface tension would allow the cleaning agents to penetrate into the dirt of the objects being cleaned (Milady, 2013). For these reasons, I predict that the laundry soaps/detergents will lower the

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surface tension of water the most and that those used for showering will lower the surface tension of water the least.

Approach to Research Question

I chose to look at how the surface tension of the water changed depending on the type of detergent in place. Many of the methods for testing the surface tension of water I encountered were complex and some required equipment that was not readily available in the school laboratory. At first, I considered using the capillary rise method to calculate the surface tension, however I went on carry out an adaptation of the Du Nuoy ring method since I thought it would be more practical. The Du Nuoy ring method is a type of experiment that allows you to

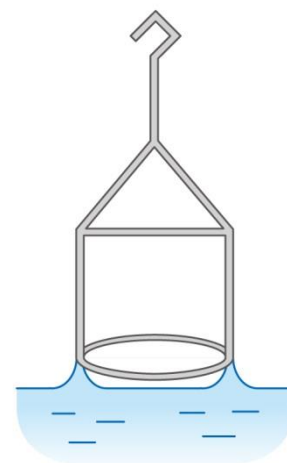


Figure 3: The Du Nuoy Ring method-
http://www.face-kyowa.co.jp/english/en_science/en_theory/en_what_Surface_tension/

calculate the surface tension of a liquid partly through the amount of force acting on the platinum ring as the ring is vertically drawn from the surface of the liquid(Kyowa Interface Science, n.d) (See Figure 3). However in my case, I used a circular metal plate since a platinum ring was not available in the school laboratory. (See Figure 4). Due to this, a limitation was that the equation for the Du Nuoy ring method would not work properly for my results as I used a circle metal.

Although my experiment would not give values for surface tension, I would be able to have results that would allow me to compare the relative effect on the surface tension by each detergent. Another limitation would be that I would be measuring the amount of force it took to lift the circular metal from the water by observing the shift of the newton meter



Figure 4: My adaptation of the Du Nuoy ring experiment

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with my eye would increase the inaccuracy of my results as well as the shift would often occur very quickly and I might not be able to see the right value.

Investigation

Variables

Independent Variable: The type of the detergent

- I will use three different brands for each type of detergent that I test.

Dependent Variable: Surface Tension

- The extent to which the surface tension will be lowered will depend on the type of detergent.

Controlled Variables: Amount of detergent used, the temperature of the water, amount of water used, hardness of water, surface area of detergent used

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Controlled Variable	Reason for controlling it	How it will be controlled
Amount of Detergent Used	The amount of detergent used should be controlled because it could affect the concentration of the detergent in the water. If this amount is not controlled for each of the detergents, then it could lead to an unfair test and inaccurate results as some detergents would be disadvantaged or advantaged depending on how much more or less detergent was put compared to the others.	I will use 50 ml of each detergent.
Amount of Water Used	The amount of water used should be controlled as it could affect the dilution of the detergent. If it is not controlled, it would lead to unfair and inaccurate results as some detergents would have an advantage as they would have a higher concentration compared to the others.	I will use 2500 ml of water in each test

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Water Hardness	Water hardness is caused by dissolved calcium and magnesium ions in the water (BBC, n.d). Non-ionic surfactants do not interact with ions in hard water (CIEC Promoting Science, 2013). However, other surfactants may be affected, if it is not controlled it could lead to inaccurate results.	I will use water from the same source for each of the test so the water hardness will remain the same throughout.
The temperature of the water	The temperature of water prior to the experiment should be controlled as it could affect the rate at which the detergents dissolve. If this is not controlled it could lead to an unfair test.	I will make sure the temperature of water in each of the tests is 20°. (Room Temperature)
The surface area of the soap/detergent	The surface area of the soap/detergent should be controlled to maintain a fair test.	All the soap/detergents utilised in this experiment will be in liquid form.

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Apparatus

- Newton Meter
- 50ml of 3 different brands of laundry soaps/detergents, dish washing soaps/detergents and showering soaps/ detergents
- 3 similar Trays
- Flat circle piece of metal
- Detachable wall hook
- Thermometer
- Weighing Scale
- Weighing Caps
- 30cm Ruler
- Volumetric Flask(2000ml)
- Measuring Cylinder(500ml)
- 3 Measuring Cylinder (50ml)

Safety

- Make sure you are wearing protective glasses as soap could enter your eyes and hurt you.
- Be sure not to swing the Newton Meter around as it has a sharp hook that could hurt you or those around you.
- Make sure you are wearing a lab coat to prevent any of the solutions or water from spilling onto your clothes

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Method

- Before you begin, make sure you have all the materials listed above. In addition, it is important that you find the centre of the of the circle metal so that it can sit on the surface of the water.
- Measure 2500 ml of water using the volumetric flask and measuring cylinder and pour it into the water basin
- Measure the temperature of the water to make sure it is 20°C (Room Temperature)
- Take 50ml of your first brand of detergent and place it into the water.
- Stir the water to make sure the detergent has dissolved into the water.
- Take your ruler and measure the midpoint of the width and height of the square and mark the point.
- Take the detachable wall hook and place it at that point.
- Now attach the hook of the Newton meter to the hook on the metal and place it in the water basin.
- Lift the metal with the Newton meter so that it is sitting on the surface.
- Gently pull the metal from the surface of the water while carefully observing the Newton meter for a reading and record the maximum reading as the metal is pulled from the surface of the water.
- Repeat this process for all three brands of each type of detergent and carry out six trials for each model tested.

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Data Analysis

Raw Data

Table showing how the force required to lift the metal from the surface of water changed with the detergent in place

		Force required to lift the metal from surface of water						
		Trials (Newtons) ± 0.05						
Detergent Type	Detergent Brand	1	2	3	4	5	6	Average Force ± 0.10 (Newtons)
Laundry Detergents	Persil	1.2	1.3	1.2	1.2	1.2	1.3	1.23
	Astonish	1.2	1.2	1.2	1.2	1.2	1.3	1.22
	Powerboy	1.1	1.1	1.1	1.2	1.1	1.1	1.12
Dish-washing detergents	Ushindi	1.5	1.5	1.5	1.4	1.4	1.5	1.47
	Sunlight	1.4	1.4	1.3	1.4	1.4	1.3	1.36
	Morning Fresh	1.4	1.3	1.4	1.4	1.5	1.4	1.40
Showering soaps	Cussons	1.6	1.6	1.5	1.6	1.6	1.6	1.58
	Johnsons	1.8	1.8	1.7	1.6	1.6	1.6	1.68
	Essential	1.6	1.6	1.7	1.6	1.7	1.7	1.65

*Average force required to dislodge the metal from the **surface of water without detergent**

= 2.07N $\pm$ 0.05

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Calculations and Assumptions

To calculate the surface tension using the Du Nouy ring method the formula is:

$$= \frac{F}{4\pi r} * C$$
 (Kyowa Interface Science, n.d.) . F can be considered to be the maximum force acting on the ring as it is vertically lifted from the surface of the liquid (Kyowa Interface Science, n.d.).

As mentioned previously the Du Nouy ring method involves the utilisation of a platinum ring. However, I utilised a circular metal instead of a platinum ring therefore the equation would not fit the method I utilised properly. Nevertheless, this equation is important to my experiment as it illustrates that the surface tension is directly proportional to F . The Force acting on the ring as it is pulled from the surface of the water arises from surface tension of the liquid (Kyowa Interface Science, n.d.).

In my results, this can be interpreted that the lower the force required to lift the circular metal plate from the surface of the water, the lower the surface tension. This would be because that all the other variables are assumed to be constant and I assume that the lower force acting on the metal would result in a lower force required to lift the metal which would result in a lower surface tension according to the equation.

I calculated the average of three brands of each type of detergent that I tested and then subtracted the weight of the circular metal plate (1N) from the values to calculate the force acting on the metal.

For example, to calculate the average amount of force required to lift the circle metal from the surface of water for laundry detergents:
$$= \frac{1.23+1.22+1.12}{3} = 1.19$$

The net force required to lift circular metal plate from surface of water would be: $1.19 - 1.00 = 0.19$

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Processed Data

Table showing how the net force required to lift the circle metal from the water changed with the detergent in place.

Detergent/Soap	Force (N) ± 0.10
Laundry Detergents	0.19
Hand Dish washing Detergents	0.41
Showering Soaps	0.64

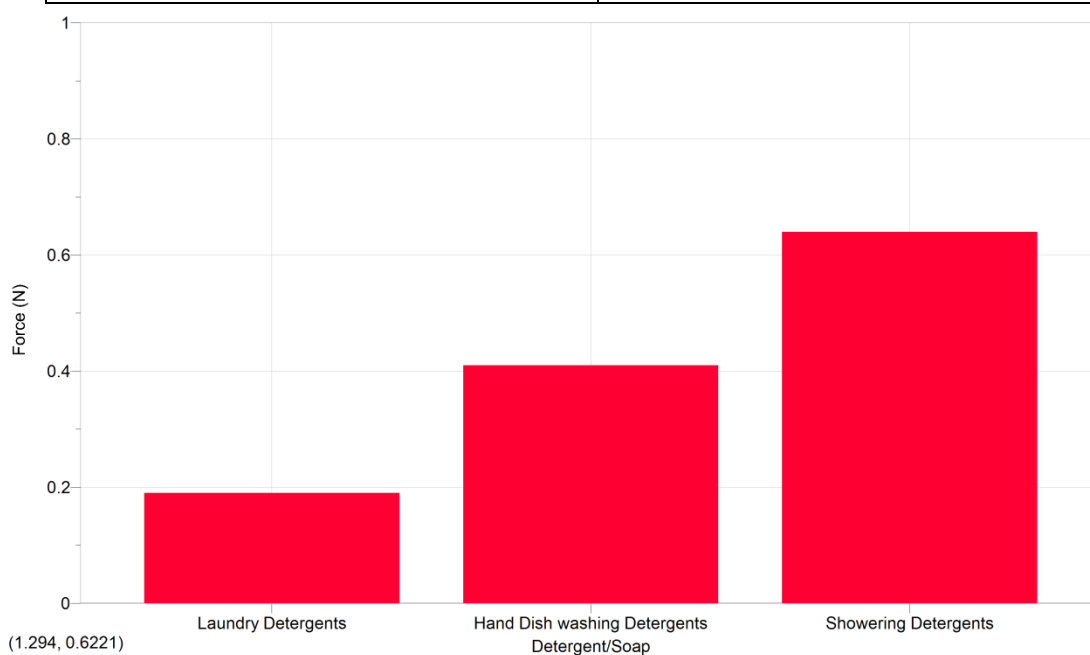


Figure 5: Bar Graph showing how the net force required to lift the metal from water changed with the type of detergent –graph made using logger pro®

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Analysis

My results show that the Laundry Detergents lowered the surface tension of the water the most compared to the other soaps/detergents. This is evident in the amount of force required to lift the circular metal from the surface of the water. The dish-washing detergent lowered the surface tension the second most and the showering detergent lowered the surface tension of the water the least based on the amount of force required to lift the circular metal from the surface of the water.

As mentioned previously, Laundry Detergents can incorporate anionic and non-ionic surfactants (American Cleaning Institute “Soaps & Detergents: Surfactants & Builders”, n.d) (CIEC Promoting Science, 2013). This would support the idea that there is a link between the cleaning ability of a detergent and how much it lowers the surface tension of water. It can be argued that laundry detergents have the strongest cleaning ability, followed by dish washing detergents and then followed by showering soaps/detergents because of the objects that they are able to clean. I believe that showering detergents are usually mild in ability as they work on the skin which is delicate, so the soaps used on the skin may be made to be mild. The surfactants that are often used for showering soaps/detergents are amphoteric surfactants which are mild (CIEC Promoting Science, 2013). On the other hand, Laundry detergents deal with strong stains, which would imply that they have a stronger cleaning ability than showering soaps/detergents.

One rule that could explain these results is Traube’s rule which shows that in a homologous series of a surfactant, the molar concentrations required to lower the surface tension to the same level decreases threefold with an additional CH_2 molecule (Atwood and Florence, 2008). A limitation to this rule could be that the rule works for the homologous series of a particular surfactant; therefore it may not be accurate to compare it with other surfactants. In

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addition, detergents/soaps containing non-ionic surfactants may have an advantage. This is because an advantage non-ionic surfactants have is that they don't interact with ions in hard water (CIEC Promoting Science, 2013).

Conclusion and Evaluation

I believe that my results proved my hypothesis was correct given that the laundry detergents lowered the surface tension of water the most with 0.19N required to pull circle metal from the surface of water while the showering detergents lowered the surface tension of water the least with a value of 0.64N required to pull the circle metal of the surface of the water. This can be attributed to the fact that laundry detergents may have a stronger cleaning ability and also contain non-ionic surfactants which are more surface active than anionic surfactants.

I believe that my experiment yielded meaningful results as although I was not able to calculate the actual surface tension values. The results I obtained allowed me to compare the relative effect of the detergents on the surface tension to a good degree. The experiment was also fair during each of the test and was done to the highest degree of care. The controlled variables were maintained throughout the experiments.

A flaw in my investigation could be that I was making assumptions on big groups of detergents based on experiments done on only three brands of each type of detergent while there could be hundreds of brands for each type of detergent currently existing. An improvement to this would be to incorporate more brands of detergent to have a stronger basis in determining which type of detergent lower the surface tension of water the most. On the other hand, when observing the three different brands used for the three types of detergents attained similar values to each other in their respective groups therefore to an extent it could be plausible that this trend continues across the groups of the detergents.

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Another challenge in this investigation would be that it is slightly difficult to determine true reason as to why laundry detergents lower the surface tension of water more compared to the other types of detergents. It could be because of the combination of the surfactants that they utilise and that they have to lower the surface tension more since they are dealing with tougher stains compared to the other types of detergents. However, there is also the possibility that the proportion of surfactants in laundry detergents is more compared to the proportion of surfactants in the other two groups of surfactants resulting in the lower surface tension. It is difficult because the formulas of the compositions of these surfactants are not usually made public.

In addition, my results were not precise as I had a large uncertainty of $\pm 0.10\text{N}$. This can be attributed to my method of experimentation, as although I tried to be as careful when measuring the values of force when lifting the ring from the surface of the water, it was difficult to be sure as it occurred very quickly. However, if I utilised a newton meter that was more sensitive, this problem could have been alleviated.

The assumptions I made on the composition of the composition of the three types of detergent were vital to my predictions and my investigation. I believe that the sources I utilised to get the surfactants that incorporate these detergents were credible. I used websites by the CIEC Promoting Science and American Cleaning Institute. I believe that these sources are credible. In addition to this, I also had first-hand information from observing the ingredients labels on each of the soaps on the types of surfactants that were used. Although many did not state the specific molecules that were used in that detergent, they usually indicated that the surfactants utilised were anionic for hand-dishwashing detergents and some indicated both anionic and non-ionic for laundry detergents. Among the showering detergents that I used, it was not indicated that amphoteric surfactants were used, therefore my assumptions in that aspect could be less reliable than for the other detergents.

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